CONCEPTUAL SERIES
Cypress

Name: ________ Cohort/Batch: ________ Date: ________ Score: ________

CYPRESS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Testing

TOTAL: 10 QUESTIONS

Q1 What is Cypress and what problem in Testing does it solve? *Threat Model*

Q6 How does Cypress handle reliability and high availability? *Observability*

Q2 What are the core components or concepts in Cypress? *Architecture*

Q7 What observability does Cypress provide out of the box? *Lifecycle*

Q3 How does Cypress compare to its main alternatives? *Configuration*

Q8 What are common performance bottlenecks in Cypress? *Compliance*

Q4 How do you get started with Cypress for a new project? *Hardening*

Q9 What security best practices apply when running Cypress? *Testing*

Q5 What configuration options most affect Cypress's behavior in production? *SSO / IdP*

Q10 What mistakes do teams commonly make when adopting Cypress? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Cypress is a tool or platform that

Mitigation

Cypress is a tool or platform that automates or simplifies a specific concern in the Testing domain, reducing manual effort and operational complexity for engineering teams.

A2 Cypress has key building blocks that work

Primitives

Cypress has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 Cypress trades off simplicity against feature richness

Hardening

Cypress trades off simplicity against feature richness differently than its alternatives. Choose Cypress when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install Cypress via its package manager or

Gotchas

Install Cypress via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 Cypress provides HA through leader election, replication,

Observability

Cypress provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 Cypress exposes metrics (Prometheus endpoint or built-in

Automation

Cypress exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run Cypress with the principle of least

Assessment

Run Cypress with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.